

ABSTRACT

With the rise of civilization, traditional open-field agriculture is facing significant challenges, particularly the decline in per capita land availability. In 1960, the global population was 3 billion, and each person had about 0.5 hectares of land. Today, with 6 billion people, that number has dropped to 0.25 hectares, and by 2050, it's projected to fall further to 0.16 hectares. Urbanization, industrialization, and global warming (leading to melting icebergs) are further reducing the amount of arable land. Soil fertility has stagnated, with no noticeable increase in productivity despite more fertilizer use. Now a days farmers are facing so many problems in agriculture field. We are trying to rectify some of the problems by our automated hydroponics system. The problems faced by farmers are small and fragmented landholdings, usage of large number of manures, Fertilizers, Biocides, Irrigation, Lack of labours, Pest control, Seasonal crops. To overcome the above problems, we are going to study which medium is suitable for plat growth except soil and which wavelength of light is suitable to give high yield. From the result of this research, we are going to implement the effective hydroponics system.

Hydroponics is a technology for growing plants in nutrient solutions (water and fertilizers) with or without the use of artificial medium (e.g., sand, gravel, vermiculite, rock wool, peat, coir, sawdust) to provide mechanical support. Poor soil health in some regions, continuous cultivation preventing natural fertility recovery, unpredictable climate conditions, rising temperatures, river pollution, poor water management, and groundwater depletion are all threatening food production in traditional soil-based farming. Given these challenges, it will likely become impossible to feed the growing population through open-field agriculture alone. As a result, soil-less cultivation, where plants grow without soil, is becoming increasingly relevant. This method, which conserves both space and water, has shown promising results worldwide. Different methods are available in hydroponics method. In our project we are using Nutrient Flow Technique (NFT), which is very easy to implement and produce high yield in small area. Keywords: Temperature Sensor, pH Sensor, Arduino, PVC pipe, Water Motor.

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ABBREVIATIONS AND ACRONYMS

Abbreviation	Full Form
IoT	Internet of Things
TDS	Total Dissolved Solids
pH	Potential of Hydrogen
EC	Electrical Conductivity
LED	Light Emitting Diode
NFT	Nutrient Film Technique
ESP8266	Espressif Systems Microcontroller 8266
ppm	Parts Per Million
RH	Relative Humidity
°C	Degree Celsius
DHT	Digital Humidity and Temperature sensor
PVC	Polyvinyl Chloride
RO	Reverse Osmosis

Chapter 1

Introduction

1.1 Introduction and Motivation

Soil is traditionally the most accessible medium for plant growth, providing essential elements like support, nutrients, air, and water. However, it also presents challenges, such as diseases, nematodes, improper pH levels, soil compaction, poor drainage, and erosion. Additionally, conventional soil-based agriculture (open field farming) demands significant space, labour, and water. In urban areas, especially metropolitan regions, soil may not be available for farming, and in some regions, fertile land is scarce due to unfavourable geography or topography. Labor shortages further complicate traditional farming practices.

Given these challenges, soil-less cultivation methods, particularly hydroponics and aeroponics, have gained traction. Hydroponics, derived from the Greek words "hydro" (water) and "ponos" (labour), involves growing plants in nutrient-rich solutions without soil. Plants can grow with their roots directly in the solution or in an inert medium like perlite or gravel. This system helps manage resources efficiently, address climate change impacts, and combat malnutrition. Aeroponics is similar to hydroponics, but plants are grown with their roots suspended in the air and misted with nutrient solutions.

In India, hydroponics was first introduced in 1946 by English scientist W. J. Shalto Douglas, who established a laboratory in Kalimpong, West Bengal, and authored the book *Hydroponics: The Bengal System*. Since then, commercial hydroponics farms have spread worldwide, with technological advancements leading to automated systems by the 1980s and home kits becoming popular in the 1990s.

We chose the "Smart Hydroponics" project because it tackles key challenges in modern agriculture, such as optimizing water use and maximizing space, while promoting sustainable food production. By incorporating smart technology, our system precisely tracks nutrient levels and water consumption, ensuring plants receive the necessary resources for optimal growth. Additionally, the integration of a camera module to monitor plant health through leaf analysis provides real-time insights, enabling proactive care. This innovative approach not only improves efficiency but also contributes to the future of sustainable farming practices.

The plant will be grown in a coco coir medium. Coco peat or coco coir plays a main role in the plant growth. Coir is a 100% natural grow and flowering medium. Coco coir will absorb the water and maintains its humidity for some time than normal soil plant. So the usage of water will be reduced by this method. The PVC pipes may be arranged in one plane or in zigzag shape depending on the types of crops grown. Many methods are available for hydroponic plant growth but here we are using Deep Flow Technique. In our project we are using many sensors. In this project water will be recycled so this type of plant growth method is very useful in summer times.

1.2 Literature Review

1. Hydroponics: An upcoming and innovative way of future farming

Hydroponics as a viable solution to global agricultural challenges caused by rapid urbanization and declining arable land. It discusses how traditional soil-based farming is becoming unsustainable due to issues like soil degradation, compaction, and poor water management. Hydroponics, a soil-less cultivation method, offers an alternative by utilizing nutrient-rich water or inert media for plant growth. The paper highlights the benefits of hydroponics, such as higher yields, reduced water usage, and the ability to grow crops in vertically stacked layers. It also outlines its historical evolution from W.F. Gericke's early experiments to modern-day applications in NASA and automated systems.

2. Smart Control Models Used for Nutrient Management in Hydroponic Crops

It classifies and evaluates technologies like artificial intelligence (AI), fuzzy logic, and artificial vision, focusing on their role in automating nutrient delivery based on plant needs. Using PRISMA guidelines, 167 articles were filtered, focusing on studies from 2020 to 2024. The paper outlines how these technologies adjust nutrient concentration in real time by interpreting data from sensors that monitor pH, electrical conductivity, and environmental conditions. The study identifies key challenges such as sensor accuracy, data volume, and crop-specific adaptations but emphasizes how AI-driven systems boost productivity and sustainability.

3. *IoT-Based Smart Precision Farming in Soilless Agriculture*

The study emphasizes how real-time monitoring, data analytics, and automation can optimize environmental conditions and nutrient management, leading to higher yields and more efficient use of resources. The authors discuss the benefits of smart farming in urban areas, showing how vertical farming and Controlled Environment Agriculture (CEA) can help address land scarcity and climate challenges. However, challenges such as high setup costs, energy consumption, and technological dependency remain key barriers.

4. *Hydroponics: Current Trends in Sustainable Crop Production*

The evolution of hydroponics from ancient practices to modern smart farming systems. Emphasis is placed on nutrient delivery optimization, reduced water use (up to 90% savings), and better

pest control without heavy chemical use. The review also explores innovations like remote cultivation. These technologies enable real-time control of temperature, humidity, and nutrient levels. Vertical farming and artificial lighting are noted as key trends that enhance crop density and year-round production. The long-term financial benefits, despite high initial investment, due to greater yields and ROI. This makes it a valuable reference for developers of smart hydroponic systems.

1. *Smart Control Models for Nutrient Management in Hydroponics*

It identifies intelligent nutrient dosing as essential for ensuring plant health and efficient resource use. Using PRISMA methodology, the authors review models that automate nutrient regulation, especially in lettuce farming. Techniques like machine learning, neural networks, and artificial vision are explored for their role in adapting nutrient delivery to plant growth stage. It also underscores the importance of real-time monitoring of pH, EC, and temperature to optimize plant growth. Fuzzy logic is praised for handling uncertain environmental data, while AI is noted for improving prediction accuracy and designing autonomous farming systems, offering insights into the effectiveness and limitations of modern nutrient control models.

2. *Lettuce Deficiency Detection Using Ensemble Multi-Dilated Adaptive Networks*

The model, called E-MD-ANet, uses a combination of CNN, VGG16, Residual Attention Networks, and LSTM to extract visual features from leaf images and classify deficiencies related to nitrogen, phosphorus, and potassium. The model is fine-tuned using the Improvised Honey Badger Algorithm (I-HBA), which improves accuracy, memory efficiency, and energy use with up to 96% accuracy in real-time classification. The framework is designed to be compatible with embedded devices, making it suitable for real-time use in IoT-based hydroponic farms. This approach not only reduces labor but also supports early detection of plant stress, improving productivity.

1.3 Objective

The objective of our "Smart Hydroponics" project is to develop a system that optimizes plant growth by delivering precise amounts of nutrients and water without the need for soil. We aim to implement a camera-based monitoring system that continuously tracks plant health by analysing leaf conditions, allowing for early detection of any potential problems. By promoting efficient water use and minimizing nutrient waste, the project focuses on resource efficiency. Additionally, we strive to create a user-friendly interface that enables easy monitoring and control of the system, making it accessible to a wide range of users. Ultimately, the goal is to contribute to sustainable and eco-friendly agriculture by providing a scalable, soil-less solution suitable for various environments, from urban farms to larger agricultural setups.

1.4 Scope of Project

Hydroponics is rapidly becoming one of the most important innovations in agriculture and could dominate food production in the future. As the global population grows and available farmland decreases due to poor land management and urban expansion, people are looking for new ways to grow food. Technologies like hydroponics and aeroponics offer exciting solutions. In cities like Tokyo, where land is scarce, hydroponic systems are already being used to grow crops like rice in underground vaults without soil. By carefully controlling the environment, they're able to harvest multiple times a year—something traditional farming can't do.

Israel, with its dry climate, has also embraced hydroponics. Companies like Organ tech use shipping containers outfitted with hydroponic systems to grow crops like berries and citrus fruits—plants that wouldn't naturally thrive in such a harsh environment. These systems can produce yields up to 1,000 times greater than what the same land area could produce using conventional farming methods, all while being fully automated.

Hydroponics is also a promising option for water-scarce regions like Africa and parts of Asia, where both food and water supplies are limited. While the initial setup cost can be high, technological advancements will eventually make it more affordable and accessible. In the long run, this method could help feed millions in areas where traditional farming isn't feasible.

Looking beyond Earth, hydroponics is set to play a key role in space exploration. NASA is researching how to use it for future missions and the possible colonization of planets like Mars or the Moon. Since there's no suitable soil in space, hydroponics could provide astronauts with a steady food supply. Not only does it offer a way to grow a variety of crops, but it also creates a bio-regenerative system—plants naturally absorb carbon dioxide and release oxygen, which is crucial for long-term space missions.

Our system's ability to automatically manage nutrients and water, monitor plant health, and optimize resources makes it perfect for both urban farming in space-constrained areas and sustainable agriculture in regions with limited water. By combining smart technology with innovative farming methods, our project is positioned to address many of the agricultural challenges of tomorrow—whether on Earth or even in space.

1.5 Organization of Report

Chapter 1: Introduction

- Explains the need for soil-less farming due to urbanization and climate challenges.
- Introduces hydroponics and the project's goal to automate it using IoT.

Chapter 2: Fundamentals

- Covers basic concepts of hydroponics and smart agriculture.
- Explains the role of sensors and IoT in monitoring plant health.

Chapter 3: Test Sites and Datasets

- Fenugreek was grown using the NFT hydroponic method.
- Real-time sensor data was collected over 15 days for analysis.

Chapter 4: Methodology

- Describes hardware (ESP8266, sensors) and software (Arduino IDE, cloud).
- Outlines workflow from system setup to data monitoring and automation

Chapter 5: Cost Analysis

- Lists all components used and their individual costs.
- Confirms the system is low-cost and scalable.

Chapter 6: Testing

- Verifies sensor accuracy by comparing with manual tools.
- Ensures all sensors perform within acceptable ranges.

Chapter 7: Results and Discussion

- Shows that fenugreek grew well under controlled conditions.
- pH, TDS, light, and temperature were key to healthy growth.

Chapter 8: Conclusion

- Summarizes successful system implementation and results.
- Highlights how the project supports sustainable agriculture.

Chapter 9: Summary

- Recaps the entire project workflow and outcomes.
- Emphasizes the potential of smart hydroponics in modern farming.

Chapter 2

Fundamentals

Fundamentals:

- The core idea of this project revolves around hydroponics, a modern farming technique where plants grow without soil—using a water-based solution rich in essential nutrients. To make this method more intelligent and efficient, we've integrated it with IoT (Internet of Things) and smart automation.
- Hydroponics Basics:
 - Plants receive nutrients directly through water, eliminating the need for soil.
 - This allows better control over nutrient intake, leading to faster growth and higher yields.
 - It's especially useful in urban or space-limited areas.
- Role of IoT:
 - IoT helps automate and remotely monitor the hydroponic system.
 - It allows real-time data collection and decision-making based on sensor inputs.
- Key Parameters Monitored:
 - pH Sensor: Ensures the nutrient solution's acidity is in the ideal range for plant growth.
 - TDS Sensor: Measures the concentration of dissolved nutrients to avoid under- or over-feeding.
 - Temperature and Humidity Sensor: Maintains the right climate conditions inside the growing environment.
 - Light Intensity Sensor: Checks if plants are getting enough light for photosynthesis.
 - Water Level Sensor: Prevents dry-out by indicating when the water/nutrient level is low.
- Microcontroller (ESP8266 NodeMCU):
 - Acts as the brain of the system.
 - Collects data from sensors and sends it to the cloud.
 - Controls actuators like pumps and valves based on sensor inputs.
- IoT Cloud Integration:

- The system is connected to an IoT cloud platform, enabling live monitoring through smartphones or computers.
- Users get alerts and insights, helping them respond quickly to any changes.
- By combining hydroponics with technology, this system not only makes farming smarter and more efficient but also accessible for students, researchers, and urban gardeners.

Chapter 3

Test Sites/Data sets

Datasets

Dataset:

- **Source:** Gathering the sensory data from the sensors and process via controller to control the water pump and share detailed data on server
- **Data Collection and storing:** The sensory data is stored in the database server and storing it for the detailed analysis and visualize for real-time monitoring.
- **Data Pre-processing:** Gathered data from different sensors like pH, TDS, Temperature and humidity, water level, water temperature and also control the water phone remotely.
- **Data Size:** In cloud each data store in JSON format using No SQL services.

Test sites:

Location: Vidyalankar Institute of Technology Wadala, Mumbai

Department: Electronics and Telecommunication Engineering

Chapter 4

Methodology

4.1 Hardware and Software Requirements

1. ESP8266 (NodeMCU):

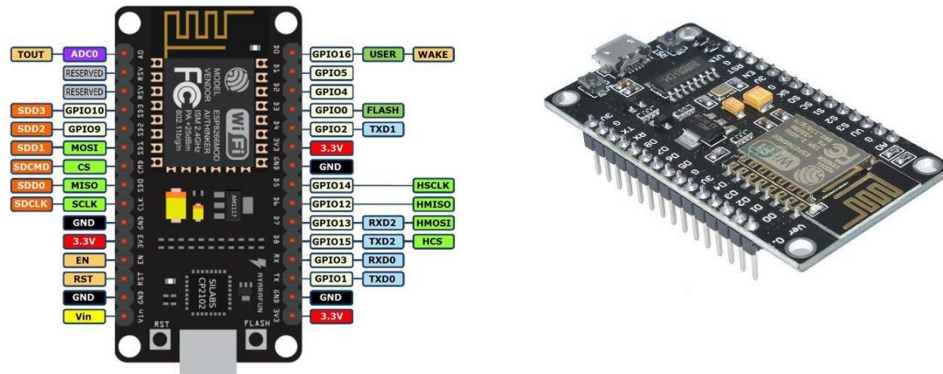


Fig 4.1.1

- Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106
- Operating Voltage: 3.3V
- Input Voltage: 7-12V
- Digital I/O Pins (DIO): 16
- Analog Input Pins (ADC): 1
- UARTs: 1
- Flash Memory: 4 MB
- SRAM: 64 KB
- Clock Speed: 80 MHz
- USB-TTL based on CP2102 is included onboard, Enabling Plug n Play
- PCB Antenna.

2. pH Sensor:



Fig 4.1.2

- input Supply voltage:- (VDC) 5
- Module Size:- (mm) 50 x 47 x 16
- Measuring Range:- 0 to 14
- PH Measuring Temperature:- 0 to 50
- Accuracy:- 0.01 pH
- pH sensor size:- (mm) 150, 12

3. TDS Sensor:

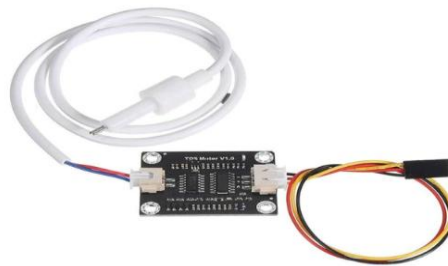


Fig 4.1.3

- Range: 0–1000 ppm (some models up to 5000 ppm).
- Analog output for nutrient concentration measurement.
- Output Voltage: $0 \sim 2.3V$
- TDS Measurement Range: $0 \sim 1000ppm$
- TDS Measurement Accuracy: $\pm 10\% FS (25\text{ }^{\circ}C)$
- Output Voltage: $0 \sim 2.3V$

4. Temperature and Humidity Sensor DHT11

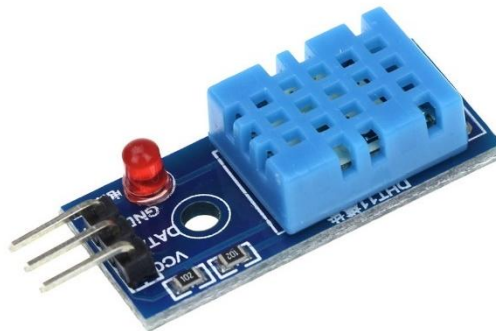


fig 4.1.4

- Temp range: 0–50°C

- Humidity rang: 20–90% R
- Digital output, powered at 3.3–5V.

5. Light Intensity Sensor (LDR Module):

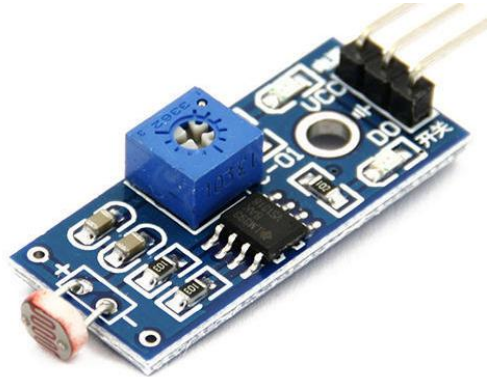


Fig 4.1.5

- Analog output based on light resistance (lower resistance in more light).
- Typically used with a 10k Ω pull-down resistor.
- Operates at 3.3–5V, simple to interface with analog pin.

6. Water Level Sensor:

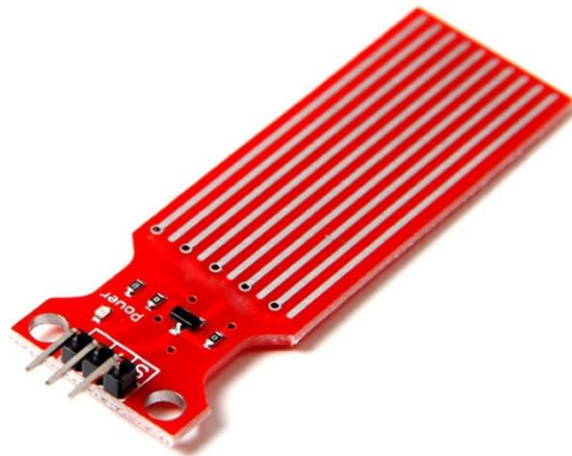


Fig 4.1.6

- Operating Voltage Range - 3 to 5 V
- Current Consumption - 3 to 41 mA
- Output Voltage Range - 0 to 3.85 V

- Sensor Output Type - Typically Analog, but possible to interface with Digital Pins
- Water Detection Area - 40mm Height x 16mm Width
- Operating Humidity - 10% to 90%
- Working Temperature - -30 to 50 °C

7. Water Pumps:



Fig 4.1.7

- DC pump, typically 5V or 12V.
- Flow rate: ~80–300 L/h (based on type).
- Used for nutrient circulation and delivery.

8. Power Supply (5V/12V):

- 5V for microcontroller and sensors.
- 12V for solenoid valves and some pumps.
- Can use adapter or battery with voltage regulator.

B. Software Requirements

- Arduino IDE: Used for programming the ESP8266 microcontroller



Fig 4.1.8

- IoT Cloud Platform (e.g., DynamoDB) For real-time monitoring and data visualization and data logging.

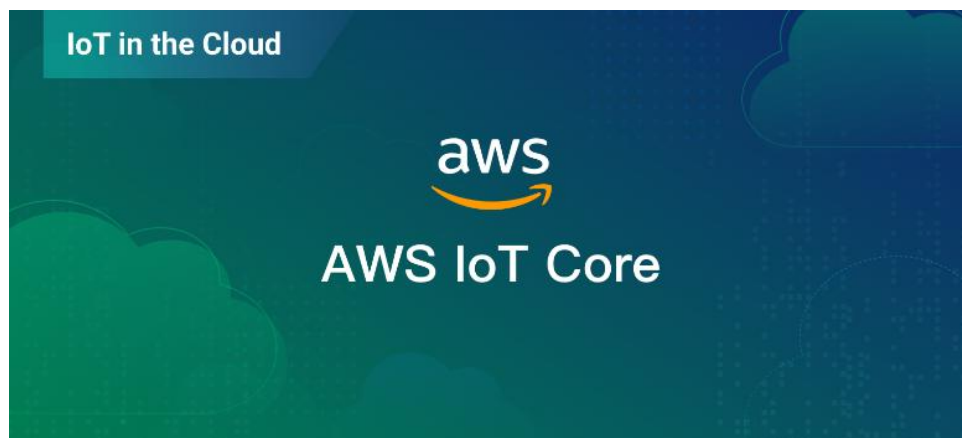


Fig 4.1.9

- Device Drivers and Libraries: Required for sensor interfacing and data handling
- Serial Monitor (Arduino IDE): For debugging and data observation during testing

4.2 Schematic diagram

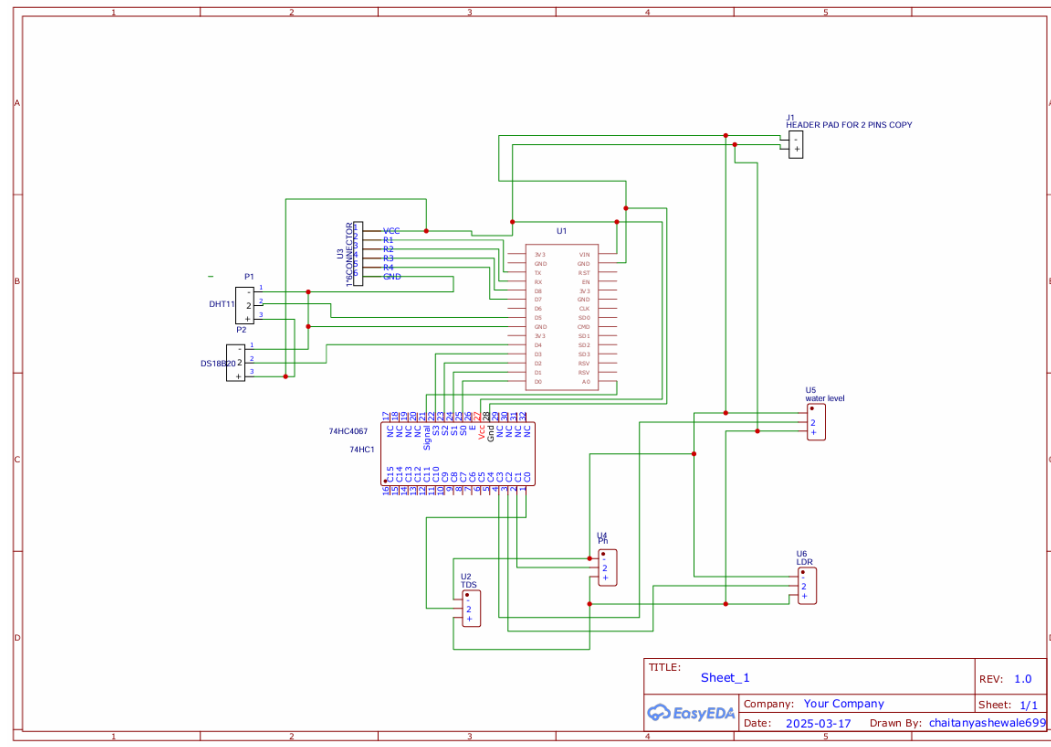


Fig 4.2.1

4.3 PCB design

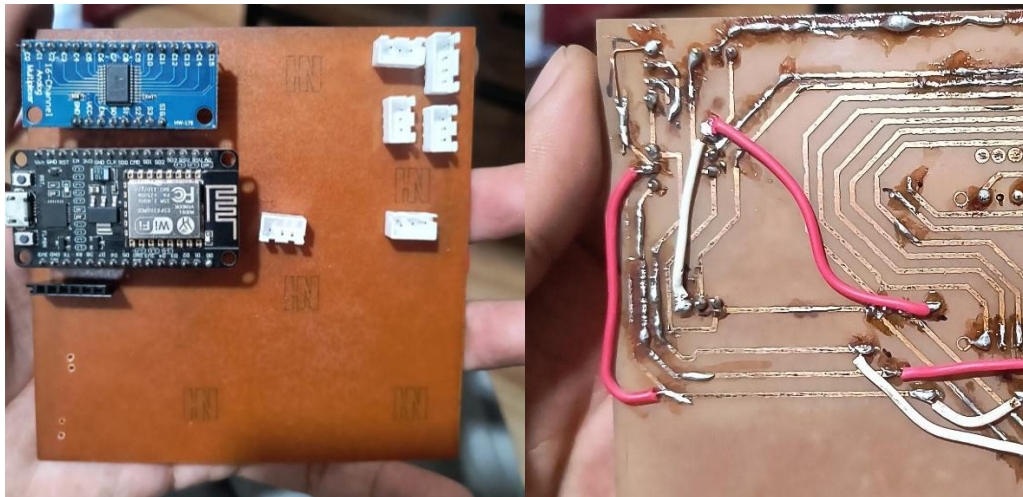


Fig 4.3.1

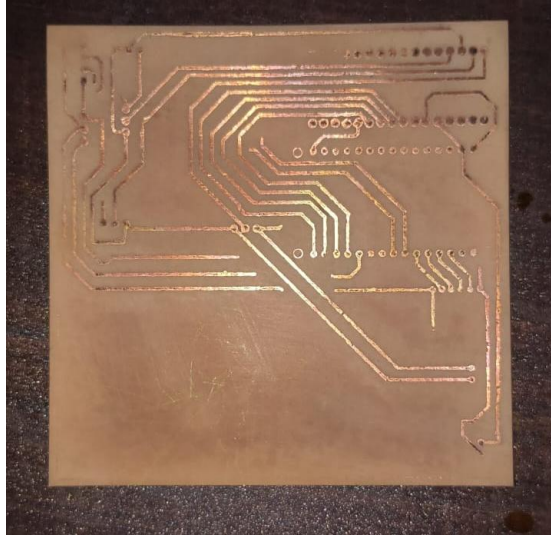


Fig 4.3.2

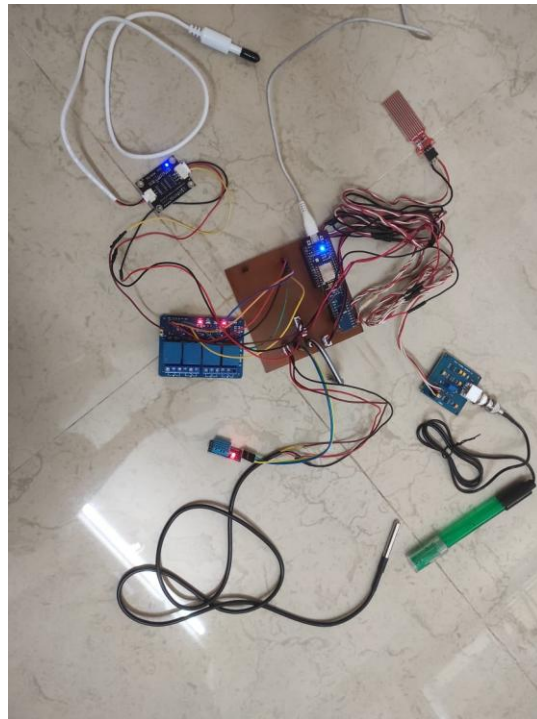


Fig 4.3.3

4.4 Workflow

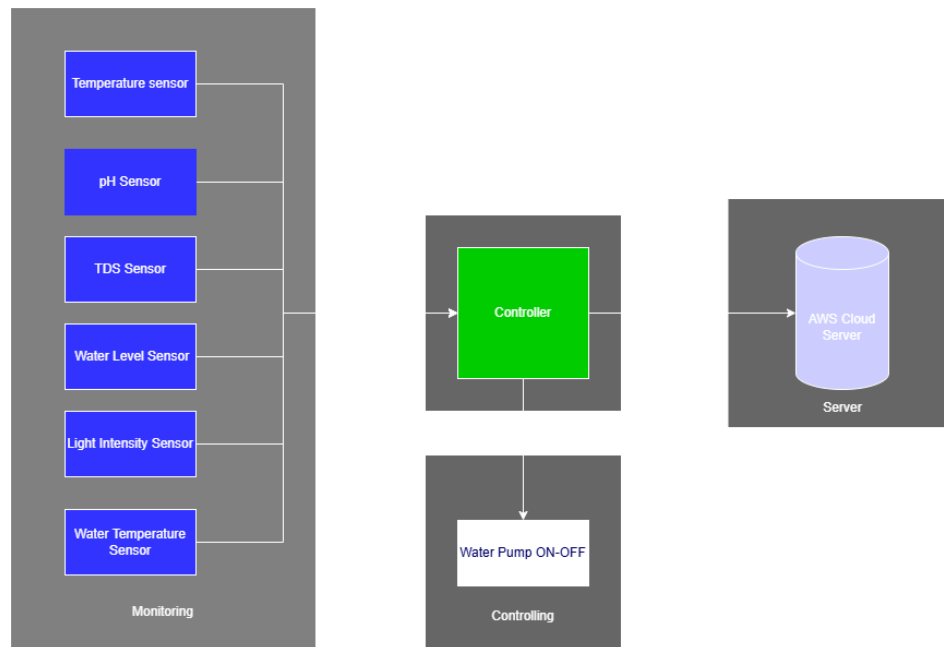


Fig 4.4.1

- **Monitoring Section**

The **monitoring block** involves collecting real-time data from various sensors. These sensors send data to the controller for processing and further action.

Sensors Used:

1. **Temperature Sensor**

Measures Surrounding temperature and humidity.

2. **pH Sensor:** Checks the acidity or alkalinity of the water solution and nutrient solution.

3. **TDS Sensor:** Measures Total Dissolved Solids, indicating nutrient concentration.

4. **Water Level Sensor;** Detects the level of water in the reservoir.

5. **Light Intensity Sensor:** Measures the amount of light available for the plants.

6. **Water Temperature Sensor:** Measures the temperature of the water to ensure it's within the optimal range.

- **Controller Section**

This is the **central processing unit** of the system. It receives data from the sensors, processes it, and takes decisions accordingly.

- **Receives Data:** from all the sensors.
- **Processes Data:** According to sensory data controller take decision and store the data in cloud server.
- **Sends Data:** to the cloud server for remote access and logging.

- **Controls Devices:** sends signals to actuators (e.g., water pump)
-
- **Controlling Section**
This block performs **actions based on sensor data** processed by the controller.
- **Water Pump ON/OFF:**
 - If the water level is low or pH/TDS needs adjustment, the controller may turn on the water pump to add or circulate nutrients.
 - This can also be scheduled or triggered remotely.
- **Server section**
AWS Cloud Server
 - Stores and logs sensor data.
 - Provides remote access to data via a web or mobile app.
 - Can be used for analytics, reporting, and long-term monitoring.

Chapter 5

Cost Analysis

Sr. No	Hardware/ Components	Quantity	Cost
1.	pH Sensor	1	1500/-
2.	DHT11 sensor	1	60/-
3.	LDR Sensor	1	40/-
4.	4 channel relay modules	1	150/-
5.	TDS sensor	1	450/-
6.	ESP 8266 (node MCU)	1	250/-
7.	74CH4067 Mux		40/-
8.	DS18B20 water temperature sensor	1	80/-
9.	Wires	-	-
10.	Male/ Female headers	10 pcs.	50/-
11.	Pipe 2.5 inch's	25 feet	500/-
12.	UPVC 1 inch's pipe	10 feet	230/-
13.	Elbow	8 pcs.	200/-
14.	Thread socket	8 pcs.	400/-
15.	End Cap 2 inches	10 pcs.	300/-
16.	Solution	1	30/-
17.	Teflon Tape	1	20/-
18.	PCB Clad	1	40/-
19.	Plywood		50/-
20.	Fertilizers		500/-
21.		Total	4,555/-

Table 5.1

Chapter 6

Testing

Objective: Test the integration of IoT sensors in a hydroponic system to monitor environmental conditions and nutrient levels for fenugreek growth.

Equipment:

Hydroponic system (NFT) with seeded fenugreek.

IoT sensors: pH sensor, TDS sensor, temperature sensor, light intensity sensor, humidity sensor.

ESP8266/IoT platform to collect and process sensor data.

Test Procedure:

pH Sensor Testing:

Test the pH sensor's accuracy by monitoring pH levels in the nutrient solution.

Cross-check with manual pH testing (using strips or a digital meter).

Expected Result: pH levels remain in the range of 6.0 to 7.0, ideal for fenugreek.

TDS Sensor Testing:

Test the TDS sensor's ability to monitor nutrient concentration.

Verify with a manual TDS meter.

Expected Result: TDS values range between 1000 to 1200 ppm for healthy fenugreek growth.

Temperature and Humidity Sensors Testing: Record ambient temperature and humidity using sensors and compare with external readings.

Expected Result: Temperature between 18°C and 24°C, humidity between 50%-70%.

Light Intensity Sensor Testing:

Measure the light intensity provided by grow lights and verify with a handheld light meter.

Expected Result: 12,000-15,000 lux to support fenugreek's photosynthesis.

Test Results:

pH sensor: Accurate within 0.1 pH of manual measurements.

TDS sensor: Reliable within 50 ppm of manual readings.

Temperature & Humidity sensors: Within $\pm 1^{\circ}\text{C}$ for temperature and $\pm 3\%$ for humidity compared to external devices.

Light sensor: Accurate within 100 lux.

Chapter 7

Results and Discussion



Fig 7.1.1

The developed IoT-based hydroponic system successfully achieved real-time monitoring and intelligent control of all critical plant growth parameters like pH ,TDS, Temperature and more important water temperature in a soilless farming environment. The sensors, including pH, TDS, water level, temperature, and light sensors, provided accurate and continuous data to the ESP8266 controller, which processed this data and acted accordingly.

Water Efficiency: Compared to traditional methods, the system used up to 80% less water through recirculation and intelligent scheduling of the water pump based on sensor input.

Cloud Monitoring & Control: Live data was successfully transmitted to the IoT dashboard, enabling users to remotely monitor conditions and control the system. Pump activation via commands was verified



Fig 7.1.2

Growth Rate:

Fenugreek plants exhibited a steady growth pattern, with germination occurring within 3-5 days, and leaf formation observed by the end of the first week. Optimal pH (6.0-7.0) and TDS (1000-1200 ppm) levels contributed to healthy nutrient absorption, while controlled temperature and humidity reduced plant stress.

Nutrient Management:

The nutrient concentration remained stable, with only minor adjustments needed during the growth cycle. The TDS sensor allowed for precise monitoring, ensuring nutrient levels were within the optimal range for fenugreek, reducing the risk of nutrient deficiencies or excesses. Continuous monitoring of pH and TDS maintained the water quality, reducing the need for frequent water changes.

1. Nutrient Profile

Nutrient	Value (ppm)	Micronutrient	Value (ppm)
Nitrogen (N)	140–200	Iron (Fe)	2–3
Phosphorus (P)	30–50	Manganese (Mn)	0.5–1
Potassium (K)	200–300	Zinc (Zn)	0.05
Calcium (Ca)	150–200	Copper (Cu)	0.05
Magnesium (Mg)	50–80	Boron (B)	0.3
Sulfur (S)	60–80	Molybdenum (Mo)	0.05
		Chlorine (Cl)	< 1

Table 7.1.1

2. Physical-Chemical Parameters

Parameter	Recommended Range
pH	5.5 – 6.5
EC (mS/cm)	1.2 – 1.8
Reservoir Temp (°C)	18 – 24
Air Temp (°C)	20 – 28
Humidity (%)	50 – 70
Light Duration	12 – 16 hours/day

Table 7.2.1

3. Nutrient Solution Maintenance

- Full solution change every 7–10 days. – Check pH & EC 1–2 times daily; adjust if pH drifts >0.2 units. – Top up with RO/distilled water to maintain EC after evaporation.

4. IoT Integration

Sensor / Actuator	Function
pH Sensor	Real-time pH monitoring
EC Sensor	Tracks nutrient concentration
Temperature Sensor	Measures water & air temperature
Humidity Sensor	Monitors ambient humidity
Water Level Sensor	Auto-controls reservoir fill
Water Pump	Timed or sensor-triggered circulation
LED Grow Lights	Scheduled or sensor-controlled

Table 7.4.1

5. Sample Nutrient Mix

Solution A (per m³): Calcium Nitrate 220g, Potassium Nitrate 160g

Solution B (per m³): Monopotassium Phosphate 50g, Magnesium Sulfate 100g, Trace mix 3g

6. Organic Fertilizer

Source	Key Nutrients
Vermicompost Tea	N, P, K + microbes
Fish Emulsion	N-rich, some P & K
Seaweed Extract	K, micronutrients
Guano Tea	High P, moderate N & K
Alfalfa Meal Tea	N & trace elements

Table 7.6.1

7. Fenugreek Sensor Data (Last 15 Days)

Date	pH	TDS (ppm)	Water Temp (°C)	Humidity (%)
2025-04-04	5.87	655.0	21.6	63.3
2025-04-05	6.45	691.0	19.0	56.2
2025-04-06	6.23	757.0	18.4	60.4
2025-04-07	6.1	730.0	23.7	60.9
2025-04-08	5.66	687.0	23.8	53.7
2025-04-09	5.66	784.0	22.9	69.4
2025-04-10	5.56	642.0	19.8	65.5
2025-04-11	6.37	688.0	18.6	68.8
2025-04-12	6.1	710.0	22.1	67.9

Table 7.7.1

Chapter 8

Summary

This project aims to design and implement a smart hydroponic farming system using Internet of Things (IoT) technology, integrated with AWS cloud services for real-time monitoring and control. The system eliminates the need for soil by growing plants in a nutrient-rich water solution, making it ideal for urban and space-constrained environments. Key environmental parameters such as pH level, electrical conductivity (EC), temperature, humidity, water level, and light intensity are continuously monitored using various IoT sensors. These sensors are connected to a microcontroller (e.g., Arduino or Raspberry Pi), which collects and processes the data.

The data is then transmitted to the AWS cloud, where it is stored and visualized on a dashboard, enabling users to remotely monitor system performance and environmental conditions. Actuators like water pumps, grow lights, and fans are controlled automatically or manually through the dashboard to maintain optimal conditions for plant growth. The system enhances resource efficiency, reduces manual labor, and supports sustainable agriculture through automation and real-time insights.

Chapter 9

Conclusion

Conclusion

The hydroponics industry is expected to grow rapidly as soil-based farming becomes increasingly difficult. In a country like India, where cities are expanding and taking over farmland, soil-less farming methods like hydroponics are becoming essential. These methods will help improve crop yields and quality, ensuring food security for the country. However, for this technology to truly take off, government support and active involvement from research institutions will be key.

Our "Smart Hydroponics" project fits perfectly into this vision for the future. As urban areas grow and available farmland shrinks, our system provides a smart, efficient solution for farming in limited spaces. With features like automated water and nutrient management and real-time plant health monitoring, we're addressing the challenges of modern agriculture. By adopting innovative approaches like hydroponics, we're not only contributing to the future of farming but also playing a role in ensuring sustainable food production for generations to come.

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